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Study of variation in Protein due to infection in Ripe wild and Hybrid Tomato by *Alternaria Solani*.Ell &Mart.

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Abstract : The present investigation deals with the purpose to examine the variation in protein content after the infection of *Alternaria Solani* in wild and hybrid Tomato.

In humid condition Tomato attacked by a disease called "Early Blight Disease." Symptom shows black infected area with multiseptate conidia which is dark brown, spindle shaped and very infectious as it causes destruction of entire plant. In this research it is investigated by taking the extract of fixed volume of tomato pulp (wild & hybrid) at definite intervals to study the Nitrogen metabolism i.e., about the variation in protein and amino acids, estimated after finding out the Optical Densit.

Key words :Early Blight Disease,Conidia, Nitrogen metabolism.

INTRODUCTION

Botanical name of Tomato is *Lycopersicon esculentum* (MILL). It belongs to the family Solanaceae.Fungal infection in Tomato has arrested the interest of number of workers because of its common use. Tomato is generally cultivated in humid condition and it is generally attacked by a disease called “Early Blight Disease”.

Tomato is important because its constituents are helpful for health. As entire fruit is edible, it can be taken directly or by cooking. Besides its advantageous effect it may be harmful when it susceptible for various diseases. “Early Blight Disease” caused by the pathogen *Alternaria Solani* of form class Deuteromycetes by Jones & Grout.

This small project work shows the study of Nitrogen metabolism. i.e., about the protein and amino acid changes due to the attack of *Alternaria Solani.* Healthy Tomato inoculated by the spore of A.solani and examined at regular intervals till rotting. This will show the exact data of rate of growth of pathogen.

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MATERIALS AND METHODS

The experimental work involved a number of standard methods available in the published papers and standard books. The technique involved qualitative as well as quantitative analysis by paper chromatographic and spectrophotometric /colorimetric methods. In addition to these, common laboratory equipments have extensively used during the investigation. Various types of chemicals also used for preparing different media and chemical analysis for chromatographic and colorimetric methods.

For the growth response, the basal cultural medium is used along with ingredients for nutritional studies of *Alternaria Solani;* Potato Dextrose with Agar is used for preparing the pure culture.

The cultures were maintained on PDA slant.

(a) Pathogenesity Test:

The Pathogenesity of the fungus was carried out on healthy Tomato fruit kept in glass chambers with fluorescent tube light fitted in it. Inoculation made on the healthy Tomato which was previously surface sterilized with mercuric chloride (1:1000) solution and washed with sterile distilled water. Small pieces of mycelial mat were used along with moist sterile cotton swab.

The inoculums with wet cotton swab were adpressed on the previously peeled fruit surface with rubber band.

The inoculated fruits kept in glass chamber were enclosed in polythene bags. Within a week spots appeared on the inoculated fruits.

From this re isolations were made which gave fresh isolates of *Alternaria solani,* which was compared with the spore of *Alternaria solani* Ref No-1420.08' from the divisions of "National Center of Fungal Taxonomy". (*Ncft.)*

(b)ChromatographicStudies:

For detection of amino acids contents in the infected fruits of both wild and hybrid Tomato, two dimensional paper chromatographic techniques were applied.Whatman chromatography paper –No 1 was uniformly used throughout the investigation.

(c) Preparation of standard chromatogram:

For identification of spots appearing on the chromatograms of fruit extracts, standard chromatograms were at first prepared. Separate chromatograms for each of the amino acid were prepared at a laboratory temperature of 30 degree C +2 degree C.

A mixture of 22 amino acids were run two dimensionally in ascending manner and the *Rf* value of each of these were determined. For each, plotting of what man chromatography paper No-1 using micropipette. The solvent used in two dimension are-Phenol: water and n Butanol: Acitic Acid: water.(v/v/v). Chromatograms were dried at room temperature after the completion of the run.

The dried chromatograms were developed by spraying them with Ninhydrin solution. They were dried in Hot Air Oven for few seconds for better results. Thus by seeing the colouration and *Rf* values, amino acids were identified.

Estimation of total amino acid contents in healthy (None inoculated as Control) and inoculated fruits (infected portion and healthy portion of infected Tomato) of two cultivars.

5gms of fruit pulp were taken from ripe wild and hybrid non infected Tomato of both cultivars in separate petridishes. Again 5gms of fruit pulp taken from healthy portion of inoculated fruit as well as infected portion of inoculated fruit each of wild and hybrid cultivar, with the help of cork borer, after the inoculation of $2^{nd,}$, 4th, 6th, to 16th days, crushed thoroughly in a mortar and pestle for 10 mints. Approximately 10 ml of Alcohol was used for boiling the fruit tissue. The heating was done in the water bath. The extract in the flask was cooled in a pan of cold water. The extract was then homogenized and centrifuged. The filtrate was collected in different sterilized dried bottles as sample. Ninhydrin is used as a spraying reagent after plotting 2ml of the tissue extract on the Whatman No-1, chromatographic paper by using two dimensional chromatographic techniques.

In this way types of amino acid present in the extract were found out by the different colored spots appeared on the paper after drying in the Hot Air Oven for few seconds. Types of amino Acid present were found out by comparing with standard *Rf* values.

For quantitative analysis of protein, Biuret methods were applied. Standard solution was prepared by bovine serum Albumin. Optical density was measured & ppm found out.

RESULT AND DISCUSSION—

The variation of protein content in infected, healthy portion of the infected Tomato and complete healthy Tomato sample is shown in the graph. On the first day, the infected sample shows significant low value. However the healthy portion of the infected sample and complete healthy sample (control) show high level of protein

The level remains almost similar, but the infected sample show gradually rises. The rise in level makes the value of Protein in the three samples almost similar.

The nature of variation in the protein content of the three is identical for 14 days.

Table 1 : Calculated Protein Data (Wild)

INFECTED PORTION		HEALTHY PART OF INFECTED		NON INFECTED (CONTROL)	
Days	ppm	Days	ppm	Days	ppm
2	10	2	130	2	160
4	110	4	130	4	160
6	180	6	130	6	160
8	180	8	150	8	10
10	50	10	150	10	110
12	80	12	50	12	20
14	10	14	80	14	60

Table-2 :Calculated Protein Data (Hybrid)

INFECTED PORTION		HEALTHY PART OF INFECTED		NON INFECTED (CONTROL)	
Days	ppm	Days	ppm	Days	ppm
2	20	2	80	2	130
4	130	4	80	4	160
6	100	6	110	6	180
8	130	8	160	8	110
10	180	10	130	10	160
12	130	12	160	12	80
14	110	14	130	14	110

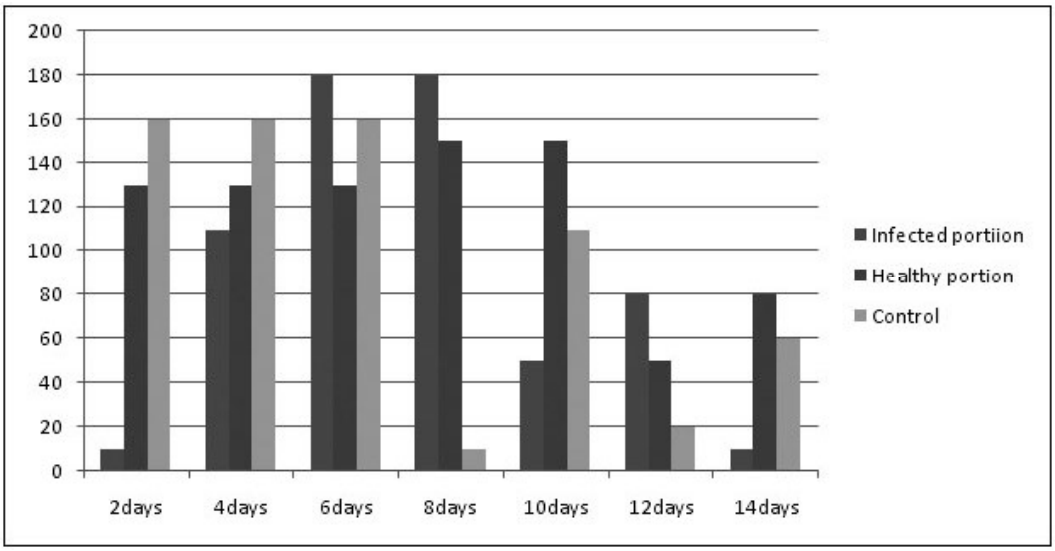


Fig.1:Column graph showing comparative protein data of Wild Tomato. (days/ppm)

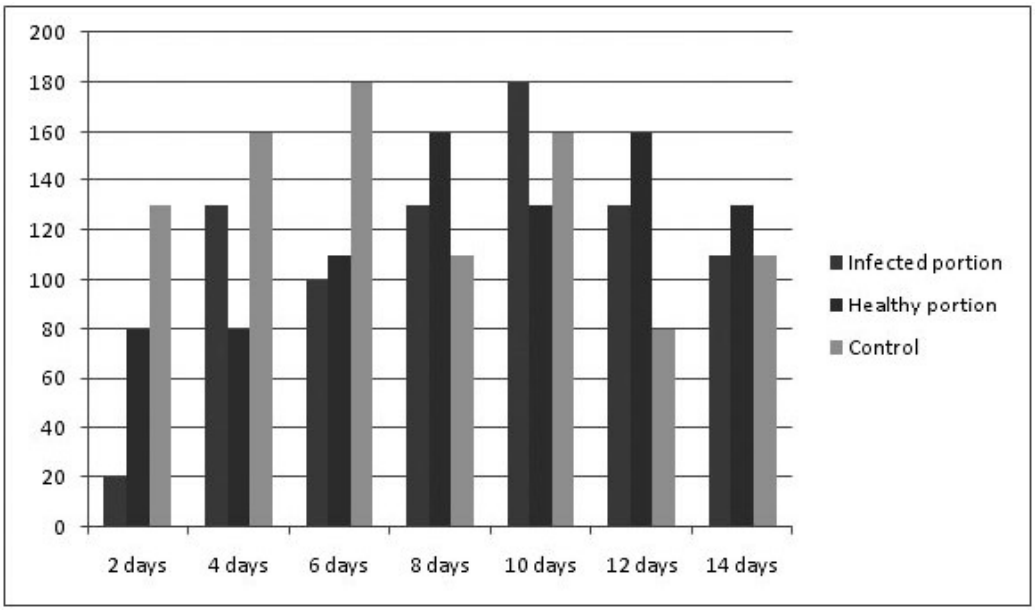


Fig.2: Column graph showing comparative protein data of Hybrid Tomato. (days/ppm)

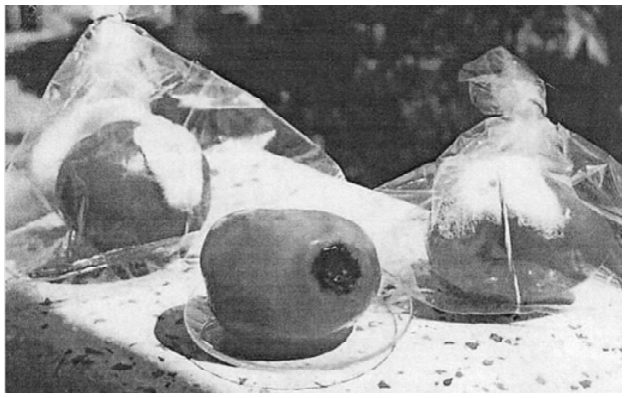


Fig.3: Inoculated Tomato

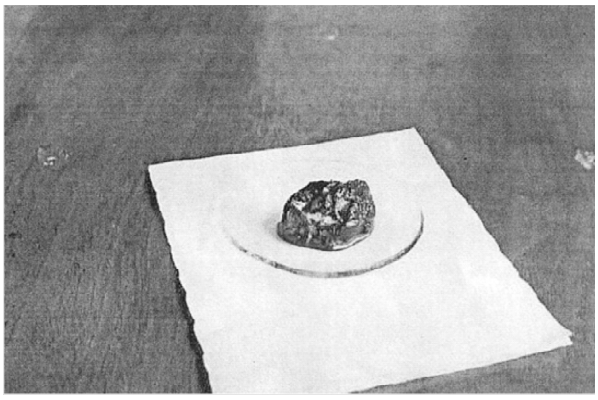


Fig.4: Fully Infected Tomato

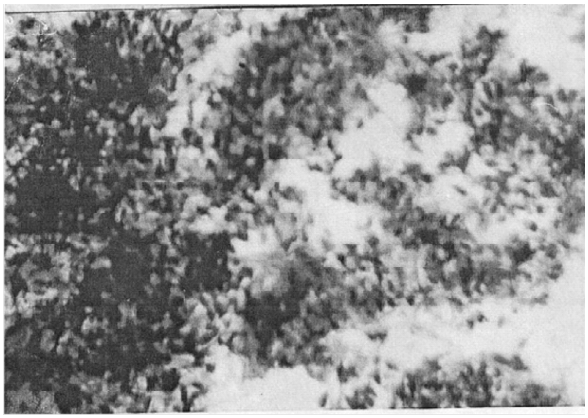


Fig.3: Conidiospores

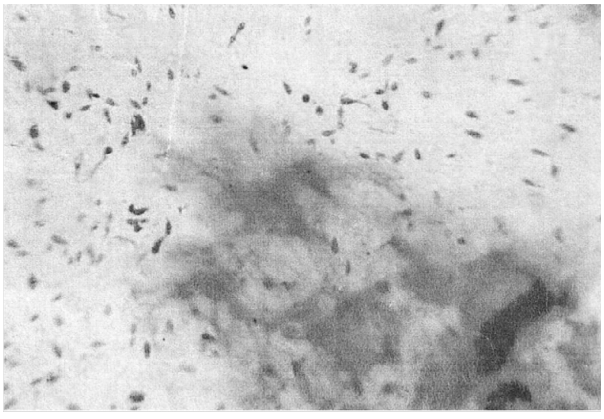


Fig.4: Conidiospores : Enlarged view

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